D3 Analytic geometry, vectors and motion: solutions

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Try each problem seriously before reading the solution. When your answer differs, find the exact step where your work and this solution part ways. All answers below were checked with computer algebra (SymPy).

Practice

1.

a. $\vec{AB} = \begin{pmatrix}3\\6\end{pmatrix}$, so the rc is $m = \frac{6}{3} = 2$. Then $y = 2x + b$ with $-1 = 4 + b$, so $b = -5$: $y = 2x - 5$.

b. $2x - y = 5$, with normal vector $\begin{pmatrix}2\\-1\end{pmatrix}$ (any nonzero multiple is also correct).

c. $\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}2\\-1\end{pmatrix} + t\begin{pmatrix}1\\2\end{pmatrix}$. Other support points and direction vectors are equally correct, e.g. $\begin{pmatrix}5\\5\end{pmatrix} + t\begin{pmatrix}3\\6\end{pmatrix}$.

2.

a. Substitute $x = 1 + 2t$, $y = 7 - t$ into $3x + 4y = 12$: $3 + 6t + 28 - 4t = 12$, so $2t = -19$ and $t = -\frac{19}{2}$. Then $x = 1 - 19 = -18$ and $y = 7 + \frac{19}{2} = 16\frac{1}{2}$. Intersection point $(-18, 16\frac{1}{2})$. Check: $3 \cdot (-18) + 4 \cdot 16\frac{1}{2} = -54 + 66 = 12$.

b. Direction vector of k: $\begin{pmatrix}4\\-3\end{pmatrix}$ (perpendicular to the normal $\begin{pmatrix}3\\4\end{pmatrix}$). Direction vector of l: $\begin{pmatrix}2\\-1\end{pmatrix}$.

$$\cos\varphi = \frac{|8 + 3|}{5 \cdot \sqrt{5}} = \frac{11}{5\sqrt{5}} \approx 0.9839, \qquad \varphi \approx 10.3^\circ.$$

3.

a. $\|\vec{a}\| = \sqrt{9 + 16} = 5$, $\|\vec{b}\| = \sqrt{4 + 25} = \sqrt{29}$. Unit vector: $\begin{pmatrix}3/5\\4/5\end{pmatrix}$. $\vec{a} + 2\vec{b} = \begin{pmatrix}3 - 4\\4 + 10\end{pmatrix} = \begin{pmatrix}-1\\14\end{pmatrix}$.

b. $\vec{a} \cdot \vec{b} = -6 + 20 = 14$. $\cos\varphi = \dfrac{14}{5\sqrt{29}} \approx 0.520$, so $\varphi \approx 58.7^\circ$. (Between vectors, no absolute value: the dot product is positive here anyway.)

c. $\begin{pmatrix}p\\2\end{pmatrix} \cdot \begin{pmatrix}3\\4\end{pmatrix} = 3p + 8 = 0$, so $p = -\frac{8}{3}$.

4. $d = \dfrac{|2 \cdot 4 - 3 - 1|}{\sqrt{4 + 1}} = \dfrac{4}{\sqrt{5}} = \frac{4}{5}\sqrt{5}$.

5. First make the coefficients equal: $6x + 8y = -1$ is the same line as $3x + 4y = -\frac{1}{2}$. Distance:

$$\frac{|7 - (-\frac{1}{2})|}{\sqrt{9+16}} = \frac{7.5}{5} = 1.5.$$

Check: $(1, 1)$ lies on the first line; its distance to $6x + 8y = -1$ is $\dfrac{|6 + 8 + 1|}{10} = 1.5$.

6.

a. $(x-3)^2 - 9 + (y+2)^2 - 4 - 12 = 0$, so $(x-3)^2 + (y+2)^2 = 25$. Centre $M(3, -2)$, radius 5.

b. $(6-3)^2 + (2+2)^2 = 9 + 16 = 25$, so T lies on c.

c. The tangent is perpendicular to $\vec{MT} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, so it is $3x+4y = c$. Through T: $c = 18+8 = 26$. Tangent: $3x + 4y = 26$.

7. $r^2 = |MA|^2 = 3^2 + 4^2 = 25$. So $(x-1)^2 + (y-2)^2 = 25$, which expands to $x^2 + y^2 - 2x - 4y - 20 = 0$.

8. Substitute: $x^2 + (x+1)^2 = 13$, so $2x^2 + 2x - 12 = 0$, i.e. $x^2 + x - 6 = (x+3)(x-2) = 0$. Then $x = 2$, $y = 3$ or $x = -3$, $y = -2$. Points $(2, 3)$ and $(-3, -2)$.

9. $x'(t) = 2t$, $y'(t) = 3t^2 - 3$.

a. $\vec{v}(2) = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$, speed $\sqrt{16+81} = \sqrt{97}$.

b. $x(t) = 3$ gives $t^2 = 4$, so $t = 2$ or $t = -2$. Now $y(2) = 8 - 6 = 2$. So at $t = 2$ the point is $(3, 2)$.

c. Horizontal: $y' = 0 \Leftrightarrow t = \pm 1$; there $x' = \pm 2 \neq 0$. Points: $t = 1$ gives $(0, -2)$, $t = -1$ gives $(0, 2)$. Vertical: $x' = 0 \Leftrightarrow t = 0$; there $y' = -3 \neq 0$. Point $(-1, 0)$.

d. Since $x(-t) = x(t)$, two different parameter values with the same x must be t and $-t$ (because $t^2 = s^2$ with $s \neq t$ forces $s = -t$). Then also $y(-t) = y(t)$, and since $y(-t) = -y(t)$ this means $y(t) = 0$: $t(t^2 - 3) = 0$, and with $t \neq 0$ we get $t = \pm\sqrt{3}$. The double point is $(3 - 1, 0) = (2, 0)$. Velocities: $\vec{v}(\pm\sqrt{3}) = \begin{pmatrix} \pm 2\sqrt{3} \\ 6 \end{pmatrix}$, each of length $\sqrt{12+36} = \sqrt{48}$.

$$\cos\varphi = \frac{|-12 + 36|}{48} = \frac{1}{2}, \qquad \varphi = 60^\circ.$$

10.

a. $(x-1)^2 + (y+2)^2 = 9\cos^2 2t + 9\sin^2 2t = 9$. So the point lies on the circle with centre $(1, -2)$ and radius 3. As t runs through an interval of length π, the angle $2t$ runs through 2π, so the whole circle is traced.

b. $\vec{v} = \begin{pmatrix} -6\sin 2t \\ 6\cos 2t \end{pmatrix}$, so $v = \sqrt{36\sin^2 2t + 36\cos^2 2t} = 6$.

c. $\vec{a} = \begin{pmatrix} -12\cos 2t \\ -12\sin 2t \end{pmatrix}$. Then $\vec{v} \cdot \vec{a} = 72\sin 2t\cos 2t - 72\cos 2t\sin 2t = 0$, so they are perpendicular. Moreover $\vec{a} = -4\begin{pmatrix} 3\cos 2t \\ 3\sin 2t \end{pmatrix} = -4\begin{pmatrix} x-1 \\ y+2 \end{pmatrix} = 4\,\vec{PM}$, where $M = (1, -2)$: a positive multiple of the vector from the point to the centre.

11.

a. $2x^2 - 1 = 2\cos^2 t - 1 = \cos 2t = y$.

b. $x = \cos t$ takes all values in $[-1, 1]$, so the arc of $y = 2x^2 - 1$ with $-1 \le x \le 1$, from $(1, 1)$ through $(0, -1)$ to $(-1, 1)$. It is traced back and forth: from $(1, 1)$ to $(-1, 1)$ for $0 \le t \le \pi$ and back for $\pi \le t \le 2\pi$.

c. $\cos t = \frac{1}{2}$ gives $t = \frac{1}{3}\pi$ or $t = \frac{5}{3}\pi$. For both, $\cos 2t = -\frac{1}{2}$ (check: $\cos \frac{2}{3}\pi = -\frac{1}{2}$ and $\cos \frac{10}{3}\pi = \cos \frac{4}{3}\pi = -\frac{1}{2}$). So $t = \frac{1}{3}\pi$ and $t = \frac{5}{3}\pi$.

CE level

12.

a. $\vec{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$, $\vec{AC} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$, $\vec{AB} \cdot \vec{AC} = 24$, $\|\vec{AB}\| = \|\vec{AC}\| = \sqrt{40}$. So $\cos \angle A = \frac{24}{40} = \frac{3}{5}$ and $\angle A \approx 53.1^\circ$.

b. A normal vector of AB is $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$, so $AB : x - 3y = c$ with $c = 1 - 3 = -2$. Distance from C:

$$\frac{|3 - 21 + 2|}{\sqrt{10}} = \frac{16}{\sqrt{10}} = \tfrac{8}{5}\sqrt{10}.$$

Area $= \frac{1}{2} \cdot |AB| \cdot d = \frac{1}{2} \cdot \sqrt{40} \cdot \dfrac{16}{\sqrt{10}} = \frac{1}{2} \cdot 2 \cdot 16 = 16$.

c. The perpendicular bisector of BC: midpoint $(5, 5)$, and $\vec{BC} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$ is a normal vector, so $-4x + 4y = 0$, i.e. $y = x$. The perpendicular bisector of AB: midpoint $(4, 2)$, normal $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$, so $3x + y = 14$. Intersecting with $y = x$: $x = y = \frac{7}{2}$. Radius squared: $|MA|^2 = (\frac{5}{2})^2 + (\frac{5}{2})^2 = \frac{25}{2}$.

$$\left(x - \tfrac{7}{2}\right)^2 + \left(y - \tfrac{7}{2}\right)^2 = \tfrac{25}{2}, \quad \text{or} \quad x^2 + y^2 - 7x - 7y + 12 = 0.$$

Check: B: $49 + 9 - 49 - 21 + 12 = 0$; C: $9 + 49 - 21 - 49 + 12 = 0$.

d. In parallelogram $ABDC$ we need $\vec{AD} = \vec{AB} + \vec{AC}$, so $D = A + \begin{pmatrix} 8 \\ 8 \end{pmatrix} = (9, 9)$. Diagonals: $\vec{AD} = \begin{pmatrix} 8 \\ 8 \end{pmatrix}$ and $\vec{BC} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$, with dot product $-32 + 32 = 0$. (This is no accident: $|AB| = |AC|$, so $ABDC$ is a rhombus; see problem 22.)

13.

a. Direction $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, so normal $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$: $x - 2y = c$ with $c = 0 - 2 = -2$. So $l : x - 2y = -2$.

b. Let $S = (0, 1)$ and $\vec{r} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Then $\vec{SP} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and the foot is $F = S + \lambda \vec{r}$ with

$$\lambda = \frac{\vec{SP} \cdot \vec{r}}{\vec{r} \cdot \vec{r}} = \frac{2 + 4}{5} = \frac{6}{5}, \qquad F = \left(\tfrac{12}{5}, \tfrac{11}{5}\right).$$

Check: $\vec{FP} = \begin{pmatrix} -7/5 \\ 14/5 \end{pmatrix}$ and $\vec{FP} \cdot \vec{r} = -\frac{14}{5} + \frac{14}{5} = 0$. Alternative: intersect l with the line through P in direction $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

c. F is the midpoint of PP', so $P' = 2F - P = \left(\frac{24}{5} - 1, \frac{22}{5} - 5\right) = \left(\frac{19}{5}, -\frac{3}{5}\right)$.

d. Method 1: $|PF| = \sqrt{\frac{49}{25} + \frac{196}{25}} = \dfrac{\sqrt{245}}{5} = \dfrac{7\sqrt{5}}{5}$. Method 2: $\dfrac{|1 - 10 + 2|}{\sqrt{5}} = \dfrac{7}{\sqrt{5}} = \frac{7}{5}\sqrt{5}$. The two agree.

14.

a. Substituting $(3, 1)$ into the left side gives $p \cdot 3 - 1 = 3p - 1$, which equals the right side for every p.

b. Write l_p as $px - y - (3p - 1) = 0$, normal $\begin{pmatrix} p \\ -1 \end{pmatrix}$. Distance from O:

$$\frac{|3p - 1|}{\sqrt{p^2 + 1}} = \sqrt{5} \iff 9p^2 - 6p + 1 = 5p^2 + 5 \iff 4p^2 - 6p - 4 = 0 \iff 2p^2 - 3p - 2 = 0.$$

So $(2p + 1)(p - 2) = 0$: $p = 2$ or $p = -\frac{1}{2}$. (Squaring was allowed because both sides are nonnegative.)

c. l_p is $y = px - 3p + 1$, with rc p. The product of the rc's is $2 \cdot (-\frac{1}{2}) = -1$, so the lines are perpendicular.

d. Every l_p passes through Q. The distance from O to a line through Q is the length of the perpendicular from O, which is at most the hypotenuse $|OQ| = \sqrt{10}$, with equality exactly when the line is perpendicular to OQ. $\vec{OQ} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ has rc $\frac{1}{3}$, so the perpendicular line has rc -3: $p = -3$. Check: $\dfrac{|-9 - 1|}{\sqrt{10}} = \sqrt{10}$. Maximum distance $\sqrt{10}$, at $p = -3$. The line through Q that is missing is the vertical line $x = 3$: every l_p has rc p, so none is vertical.

15.

a. $|MP| = \sqrt{7^2 + 1^2} = \sqrt{50} = 5\sqrt{2} > 5$.

b. Write the line as $ax - y - 9a + 2 = 0$. It touches c exactly when its distance to $M(2, 1)$ is 5:

$$\frac{|2a - 1 - 9a + 2|}{\sqrt{a^2 + 1}} = 5 \iff (1 - 7a)^2 = 25(a^2 + 1) \iff 24a^2 - 14a - 24 = 0 \iff 12a^2 - 7a - 12 = 0.$$

Discriminant $49 + 576 = 625$, so $a = \dfrac{7 \pm 25}{24}$: $a = \frac{4}{3}$ or $a = -\frac{3}{4}$.

c. For $a = \frac{4}{3}$: $y = \frac{4}{3}x - 10$, i.e. $4x - 3y = 30$. The touching point is the foot of the perpendicular from M: $M + \lambda \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ with $4(2 + 4\lambda) - 3(1 - 3\lambda) = 30$, so $25\lambda = 25$, $\lambda = 1$: $T_1 = (6, -2)$.

For $a = -\frac{3}{4}$: $3x + 4y = 35$. Then $M + \lambda \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ with $3(2 + 3\lambda) + 4(1 + 4\lambda) = 35$, so $\lambda = 1$: $T_2 = (5, 5)$.

d. $\vec{MT_1} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\vec{T_2P} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ are equal, so MT_1PT_2 is a parallelogram. $\vec{T_1P} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$; both side vectors have length 5 and $\begin{pmatrix} 4 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix} = 0$. A parallelogram with equal adjacent sides and a right angle is a square.

16.

a. Expand c_2: $x^2 + y^2 - 14x - 2y + 25 = 0$. Subtracting from $x^2 + y^2 - 25 = 0$ gives $14x + 2y - 50 =$

0, so $y = 25 - 7x$. Substitute into c_1: $x^2 + (25-7x)^2 = 25$, so $50x^2 - 350x + 600 = 0$, i.e. $x^2 - 7x + 12 = 0$, so $x = 3$ or $x = 4$. Points $S_1(3,4)$ and $S_2(4,-3)$.

b. $|S_1S_2| = \sqrt{1+49} = \sqrt{50} = 5\sqrt{2}$.

c. At S_1 the radii are $\vec{OS_1} = \begin{pmatrix}3\\4\end{pmatrix}$ and $\vec{M_2S_1} = \begin{pmatrix}-4\\3\end{pmatrix}$, with dot product $-12 + 12 = 0$. Each tangent is perpendicular to its radius, so the tangents are perpendicular too. At S_2: $\begin{pmatrix}4\\-3\end{pmatrix} \cdot \begin{pmatrix}-3\\-4\end{pmatrix} = 0$. (Or: the distance between centres is $\sqrt{50}$ and $50 = 25 + 25 = r_1^2 + r_2^2$, so by the converse of Pythagoras the radii to an intersection point are perpendicular.)

17.

a. Eliminate t. For P: $t = y - 1$, so $x = -3 + 2(y-1)$, i.e. $x - 2y = -5$. For Q: $t = 5 - x$, so $y = -2 + 3(5-x)$, i.e. $3x + y = 13$. Solving: $x = 2y - 5$ gives $6y - 15 + y = 13$, so $y = 4$, $x = 3$. Intersection $(3,4)$.

b. P is at $(3,4)$ when $-3 + 2t = 3$, so $t = 3$. Q is at $(3,4)$ when $5 - t = 3$, so $t = 2$. Different times, so no collision. (They can meet nowhere else, since the paths share only this point.)

c. Directions $\begin{pmatrix}2\\1\end{pmatrix}$ and $\begin{pmatrix}-1\\3\end{pmatrix}$: $\cos\varphi = \dfrac{|-2+3|}{\sqrt{5}\sqrt{10}} = \dfrac{1}{\sqrt{50}}$, so $\varphi \approx 81.9°$.

d. $\vec{PQ} = \begin{pmatrix}8-3t\\-3+2t\end{pmatrix}$, so

$$d(t)^2 = (8-3t)^2 + (2t-3)^2 = 13t^2 - 60t + 73.$$

This parabola has its minimum at $t = \frac{60}{26} = \frac{30}{13}$ (which is ≥ 0). There $d^2 = 13\cdot\frac{900}{169} - \frac{1800}{13} + 73 = \frac{900-1800+949}{13} = \frac{49}{13}$, so the smallest distance is $\dfrac{7}{\sqrt{13}} = \frac{7}{13}\sqrt{13}$. Positions: $P = \left(\frac{21}{13}, \frac{43}{13}\right)$, $Q = \left(\frac{35}{13}, \frac{64}{13}\right)$. Check: $\vec{PQ} = \frac{7}{13}\begin{pmatrix}2\\3\end{pmatrix}$ has length $\frac{7}{13}\sqrt{13}$, and it is perpendicular to the relative velocity $\vec{v}_Q - \vec{v}_P = \begin{pmatrix}-3\\2\end{pmatrix}$.

18. $x'(t) = \cos t$, $y'(t) = 2\cos 2t$.

a. $x = 0$ and $y = 0$ need $\sin t = 0$, so $t = 0, \pi, 2\pi$ ($t = 0$ and $t = 2\pi$ give the same passage). $\vec{v}(0) = \begin{pmatrix}1\\2\end{pmatrix}$, $\vec{v}(\pi) = \begin{pmatrix}-1\\2\end{pmatrix}$. $\cos\varphi = \dfrac{|-1+4|}{5} = \dfrac{3}{5}$, so $\varphi \approx 53.1°$.

b. Horizontal: $\cos 2t = 0$ gives $t = \frac{1}{4}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi$; there $\cos t \neq 0$. Points $(\frac{1}{2}\sqrt{2}, 1)$, $(\frac{1}{2}\sqrt{2}, -1)$, $(-\frac{1}{2}\sqrt{2}, 1)$, $(-\frac{1}{2}\sqrt{2}, -1)$. Vertical: $\cos t = 0$ gives $t = \frac{1}{2}\pi, \frac{3}{2}\pi$; there $y' = 2\cos\pi = -2$ and $2\cos 3\pi = -2$, both nonzero. Points $(1,0)$ and $(-1,0)$.

c. $y^2 = (2\sin t\cos t)^2 = 4\sin^2 t(1 - \sin^2 t) = 4x^2(1-x^2)$.

d. $\sin 2t = \sin t \Leftrightarrow \sin t(2\cos t - 1) = 0 \Leftrightarrow \sin t = 0$ or $\cos t = \frac{1}{2}$. So $t \in \{0, \frac{1}{3}\pi, \pi, \frac{5}{3}\pi, 2\pi\}$, giving the points $(0,0)$, $(\frac{1}{2}\sqrt{3}, \frac{1}{2}\sqrt{3})$ and $(-\frac{1}{2}\sqrt{3}, -\frac{1}{2}\sqrt{3})$.

e. $v^2 = \cos^2 t + 4\cos^2 2t \leq 1 + 4 = 5$, and at $t = 0$ both terms reach their maximum at the same time: $v(0)^2 = 5$. So the maximum speed is $\sqrt{5}$ (at the origin, at $t = 0$ and $t = \pi$).

19. $\vec{v} = \begin{pmatrix}1-\cos t\\ \sin t\end{pmatrix}$, $\vec{a} = \begin{pmatrix}\sin t\\ \cos t\end{pmatrix}$.

a. $v^2 = 1 - 2\cos t + \cos^2 t + \sin^2 t = 2 - 2\cos t = 4\sin^2\frac{1}{2}t$ (since $\cos t = 1 - 2\sin^2\frac{1}{2}t$). For $0 \leq t \leq 2\pi$ we have $0 \leq \frac{1}{2}t \leq \pi$, so $\sin\frac{1}{2}t \geq 0$ and $v = 2\sin\frac{1}{2}t$. Maximum 2 at $t = \pi$, at the

point $(\pi, 2)$: the top of the arch.

b. $\|\vec{a}\| = \sqrt{\sin^2 t + \cos^2 t} = 1$.

c. At $t = \frac{1}{2}\pi$: point $(\frac{1}{2}\pi - 1, 1)$, velocity $\begin{pmatrix}1\\1\end{pmatrix}$, so rc 1. Tangent: $y - 1 = x - \frac{1}{2}\pi + 1$, i.e. $y = x + 2 - \frac{1}{2}\pi$.

d. $\vec{v}(0) = \vec{v}(2\pi) = \vec{0}$: the point is momentarily at rest (the rim point touches the ground). The curve has a cusp (*keerpunt*) there: near $t = 0$ the direction $\dfrac{y'}{x'} = \dfrac{\sin t}{1 - \cos t} = \dfrac{1}{\tan \frac{1}{2}t}$ tends to $\pm\infty$, so the curve arrives and leaves vertically.

20. $\vec{v} = \begin{pmatrix}-4\sin t\\ 2\cos t\end{pmatrix}$, $\vec{a} = \begin{pmatrix}-4\cos t\\ -2\sin t\end{pmatrix}$.

a. $\dfrac{16\cos^2 t}{16} + \dfrac{4\sin^2 t}{4} = 1$.

b. $v^2 = 16\sin^2 t + 4\cos^2 t = 4 + 12\sin^2 t$. Since $0 \le \sin^2 t \le 1$: maximum speed $\sqrt{16} = 4$ when $\sin^2 t = 1$ ($t = \frac{1}{2}\pi, \frac{3}{2}\pi$), at $(0, 2)$ and $(0, -2)$; minimum speed $\sqrt{4} = 2$ when $\sin t = 0$ ($t = 0, \pi$), at $(4, 0)$ and $(-4, 0)$.

c. $\vec{a} = -\begin{pmatrix}4\cos t\\ 2\sin t\end{pmatrix} = -\vec{OP} = \vec{PO}$.

d. $\vec{OP} \cdot \vec{v} = -16\sin t\cos t + 4\sin t\cos t = -12\sin t\cos t = -6\sin 2t$. This is zero for $t = 0, \frac{1}{2}\pi, \pi, \frac{3}{2}\pi$: the points $(4, 0)$, $(0, 2)$, $(-4, 0)$, $(0, -2)$.

e. At $t = \frac{1}{3}\pi$: point $(2, \sqrt{3})$, $\vec{v} = \begin{pmatrix}-2\sqrt{3}\\ 1\end{pmatrix}$, so a normal vector is $\begin{pmatrix}1\\ 2\sqrt{3}\end{pmatrix}$. Tangent: $x + 2\sqrt{3}\,y = 2 + 6 = 8$.

Challenge

21.

a. A point (x, y) is equally far from k and l exactly when

$$\frac{|3x + 4y - 12|}{5} = \frac{|5x - 12y - 4|}{13} \iff 13(3x + 4y - 12) = \pm 5(5x - 12y - 4).$$

With $+$: $14x + 112y = 136$, i.e. $7x + 56y = 68$. With $-$: $64x - 8y = 176$, i.e. $8x - y = 22$.

b. Normals $\begin{pmatrix}7\\56\end{pmatrix}$ and $\begin{pmatrix}8\\-1\end{pmatrix}$: dot product $56 - 56 = 0$.

c. Write the lines as $\vec{u} \cdot X = c_1$ and $\vec{w} \cdot X = c_2$ with unit normals $\vec{u}$, $\vec{w}$ (divide each equation by the length of its normal). The distance formula becomes $|\vec{u} \cdot X - c_1| = |\vec{w} \cdot X - c_2|$, so the bisectors are $(\vec{u} - \vec{w}) \cdot X = c_1 - c_2$ and $(\vec{u} + \vec{w}) \cdot X = c_1 + c_2$. Since the lines intersect, $\vec{u} \neq \pm\vec{w}$, so both normals $\vec{u} \pm \vec{w}$ are nonzero and these are genuine lines. Finally $(\vec{u} + \vec{w}) \cdot (\vec{u} - \vec{w}) = \vec{u} \cdot \vec{u} - \vec{w} \cdot \vec{w} = 1 - 1 = 0$. ■

22.

a. Let the rhombus be O, A, C, B with $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, $\vec{OC} = \vec{a} + \vec{b}$, and $\|\vec{a}\| = \|\vec{b}\|$. The diagonals are $\vec{a} + \vec{b}$ and $\vec{b} - \vec{a}$, and $(\vec{a} + \vec{b}) \cdot (\vec{b} - \vec{a}) = \|\vec{b}\|^2 - \|\vec{a}\|^2 = 0$. ■

b. Let O be the centre, $\vec{OA} = \vec{a}$, $\vec{OB} = -\vec{a}$, $\vec{OC} = \vec{c}$, with $\|\vec{a}\| = \|\vec{c}\| = r$. Then $\vec{CA} \cdot \vec{CB} = (\vec{a} - \vec{c}) \cdot (-\vec{a} - \vec{c}) = \|\vec{c}\|^2 - \|\vec{a}\|^2 = 0$. Since $C \neq A, B$, both vectors are nonzero, so

$\angle ACB = 90°$. ■

c. In a parallelogram with sides $\vec{a}$, $\vec{b}$ the diagonals are $\vec{a}+\vec{b}$ and $\vec{b}-\vec{a}$. If they are perpendicular, the computation in a gives $\|\vec{b}\|^2 - \|\vec{a}\|^2 = 0$, so adjacent sides are equal and it is a rhombus. ■

23.

a. Expanding:

$$(u_1^2+u_2^2)(v_1^2+v_2^2) - (u_1v_1+u_2v_2)^2 = u_1^2v_2^2 + u_2^2v_1^2 - 2u_1v_1u_2v_2 = (u_1v_2-u_2v_1)^2 \geq 0.$$

Equality holds exactly when $u_1v_2 = u_2v_1$, i.e. when $\vec{u}$ and $\vec{v}$ are parallel (one is a multiple of the other, or one is $\vec{0}$).

b. $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + 2\,\vec{u}\cdot\vec{v} + \|\vec{v}\|^2 \leq \|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\| + \|\vec{v}\|^2 = (\|\vec{u}\|+\|\vec{v}\|)^2$, using a in the form $\vec{u}\cdot\vec{v} \leq |\vec{u}\cdot\vec{v}| \leq \|\vec{u}\|\|\vec{v}\|$. Both sides are nonnegative, so take square roots. ■

24. Both sides are nonnegative, so $|PA| = 2|PB| \Leftrightarrow |PA|^2 = 4|PB|^2$:

$$x^2+y^2 = 4\big((x-3)^2+y^2\big) \iff 3x^2+3y^2-24x+36 = 0 \iff x^2+y^2-8x+12 = 0 \iff (x-4)^2+y^2 = 4.$$

Every step is an equivalence, so the set of possible positions is exactly the circle with centre $(4,0)$ and radius 2. (Check: $(2,0)$ has $|PA|=2$, $|PB|=1$; $(6,0)$ has 6 and 3.)

25.

a. The midpoint is $(\cos t, \sin t)$, and $\cos^2 t + \sin^2 t = 1$: it moves on the unit circle (the quarter in the first quadrant).

b. $R = (2\cos t, 0) + \frac{1}{3}\big((0,2\sin t) - (2\cos t, 0)\big) = \big(\frac{4}{3}\cos t, \frac{2}{3}\sin t\big)$. Then $\dfrac{9x^2}{16} + \dfrac{9y^2}{4} = \cos^2 t + \sin^2 t = 1$.

26. The loop runs from $(2,0)$ at $t=-\sqrt{3}$ back to $(2,0)$ at $t=\sqrt{3}$, so it is one round trip.

$$\int_{-\sqrt{3}}^{\sqrt{3}} (t^3-3t)\cdot 2t\,dt = \int_{-\sqrt{3}}^{\sqrt{3}} (2t^4-6t^2)\,dt = 2\left[\tfrac{2}{5}t^5 - 2t^3\right]_0^{\sqrt{3}} = 2\left(\tfrac{18}{5}\sqrt{3} - 6\sqrt{3}\right) = -\tfrac{24}{5}\sqrt{3}.$$

(We used that the integrand is even, and $(\sqrt{3})^5 = 9\sqrt{3}$, $(\sqrt{3})^3 = 3\sqrt{3}$.) The area is $\frac{24}{5}\sqrt{3} \approx 8.31$. The minus sign only says that the loop is traversed clockwise. A numerical check with the shoelace formula on 200 000 points of the curve gives 8.3138.

27. From problem 19, $v(t) = 2\sin\frac{1}{2}t$ on $[0, 2\pi]$, so

$$L = \int_0^{2\pi} 2\sin\tfrac{1}{2}t\,dt = \left[-4\cos\tfrac{1}{2}t\right]_0^{2\pi} = 4+4 = 8.$$

28.

a. $\vec{a}\cdot\vec{b} = 2-2+4 = 4$, $\|\vec{a}\| = \sqrt{1+4+4} = 3$, $\|\vec{b}\| = \sqrt{4+1+4} = 3$. $\cos\varphi = \frac{4}{9}$, $\varphi \approx 63.6°$.

b. We need $\vec{n} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$ with $p+2q+2r=0$ and $2p-q+2r=0$. Subtracting: $-p+3q=0$, so $p=3q$; then $5q+2r=0$. Take $q=2$: $p=6$, $r=-5$. So $\vec{n} = \begin{pmatrix} 6 \\ 2 \\ -5 \end{pmatrix}$ (or any nonzero multiple). Check: $6+4-10=0$ and $12-2-10=0$. This is the cross product $\vec{a}\times\vec{b}$ you will meet in Wiskunde D.

c. $\dfrac{|0+0+0-9|}{\sqrt{1+4+4}} = \dfrac{9}{3} = 3$. Proof of the formula: let $Q(x_1, y_1, z_1)$ be any point of the plane $ax + by + cz = d$ and $\vec{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$. Exactly as in two dimensions, $\vec{n}$ is perpendicular to every vector in the plane (subtract the equations for two points of the plane), so the distance from $P(x_0, y_0, z_0)$ to the plane is the length of the projection of $\vec{QP}$ onto $\vec{n}$:

$$d = \frac{|\vec{QP} \cdot \vec{n}|}{\|\vec{n}\|} = \frac{|a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}. \qquad \blacksquare$$